I. Background Information

Sara Elizabeth Siegler, the CEO/Owner of Sara Elizabeth Siegler ("SES"), or Ohio entity number 2029865, attended both the FDA Scientific Evidence in the Development of Human Cells, Tissues, and Cellular and Tissue-Based Products Subject to Premarket Approval ("SEDHC") workshop as well as the Part 15 Public Hearing convened on the present matter (*i.e.*, Docket No. FDA-2015-D-3719).[1] For background purposes, the mission statement of the FDA is as follows:[2]

> FDA is responsible for protecting the public health by assuring the safety, efficacy and security of human and veterinary drugs, biological products, medical devices, our nation's food supply, cosmetics, and products that emit radiation.
>
> FDA is also responsible for advancing the public health by helping to speed innovations that make medicines more effective, safer, and more affordable and by helping the public get the accurate, science-based information they need to use medicines and foods to maintain and improve their health. FDA also has responsibility for regulating the manufacturing, marketing and distribution of tobacco products to protect the public health and to reduce tobacco use by minors.
>
> FDA also plays a significant role in the Nation's counterterrorism capability. FDA fulfills this responsibility by ensuring the security of the food supply and by fostering development of medical products to respond to deliberate and naturally emerging public health threats.

Within this context, the comments of SES follow.

II. Draft Guidelines & Recommendations

A. Minimal Manipulation

The idea that registration in clinicaltrials.gov and/or potentially misleading marketing that may lead patients to believe that they are enrolled on a clinical trial being conducted under an IND/IDE when that may or may not be the case arose at both the FDA SEDHC workshop as well as the Part 15 public hearing convened on the present matter (*i.e.*, Docket No. FDA-2015-D-3719). Ensuring that individuals have access to accurate information regarding stem cell therapies, whether proven or otherwise, should be a priority in terms of protecting the health and safety of the public.

While SES acknowledges that there are likely regulatory and/or legal avenues that could be undertaken to mitigate this issue, which SES views as an issue of informed consent, SES proposes revisions to the HCT/P Establishment Registration (OMB No.

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[1]See: http://www.fda.gov/downloads/BiologicsBloodVaccines/NewsEvents/WorkshopsMeetingsConferences/UCM497048.pdf

[2]Source URL: http://www.fda.gov/downloads/aboutfda/reportsmanualsforms/reports/budgetreports/ucm298331.pdf

0910-0543) such that the public could be able to easily determine the status of the HCT/P therapy since all manufacturers are required to complete said HCT/P Establishment Registration.[3,4] That is, 21 Code of Federal Regulation ("CFR") Part1271 (21 CFR 1271) requires establishments that perform one or more steps in the manufacture of HCT/Ps to register and list their products with the FDA.[5]

The proposed changes to the HCT/P Establishment Registration form are as follows:

- Expand the HCT/P establishment registration requirement to all establishments that manufacture and/or administer HCT/Ps via an amendment to 21 CFR 1271; and
- For each HCT/P type (bone, cartilage, *etc.*), incorporate a feature such that manufacturers must designate whether the product is:
  - being evaluated in a clinical trial under an IND/IDE;
  - approved for marketing by FDA; or
  - unproven (*i.e.*, not evaluated by FDA for safety and/or efficacy).

A simple, single page primer could be provided to the public by way of FDA's website (and hopefully patient advocacy groups as well), and said primer would provide the public with instructions on how to easily access information regarding a HCT/P therapy via the HCT/P Establishment Registration pubic query website (https://www.accessdata.fda.gov/scripts/cber/CFAppsPub/tiss/index.cfm) in addition to providing brief explanations of the implications of each additional proposed product designation (being evaluated in a clinical trial under an IND/IDE, approved for marketing by FDA or unproven (*i.e.*, not evaluated by FDA for safety and/or efficacy)).[6]

The Clinical Trials Registration and Results Information Submission Final Rule, pursuant to 42 CFR Part 11, was published in the Federal Register ("FR") on September 21, 2016 as Docket Number NIH-2011-0003 (FR Document Number 2016-22129; RIN: 0925-AA55),[7] and the complementing NIH policy (*NIH Policy on the Dissemination of NIH-Funded Clinical Trial Information*), which takes effect on January 18, 2017, was published in the FR as FR Document Number 2016-22379.[8] With regard to the first of the three proposed product/product candidate designations (*i.e.,* being evaluated in a clinical trial under an IND/IDE), SES asks

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[3]The source URL for the *Human Cell and Tissue Establishment Registration Public Query* website is as follows: https://www.accessdata.fda.gov/scripts/cber/CFAppsPub/tiss/index.cfm

[4]An example HUMAN CELL AND TISSUE ESTABLISHMENT REGISTRATION (HCTERS) follows in the Appendix and is incorporated here for reference.

[5]Source URL: http://www.fda.gov/ForIndustry/FDABasicsforIndustry/ucm237987.htm

[6]See the following URL: http://www.fda.gov/forconsumers/consumerupdates/ucm286155.htm

[7]Source URL: https://s3.amazonaws.com/public-inspection.federalregister.gov/2016-22129.pdf

[8]Source URL: https://s3.amazonaws.com/public-inspection.federalregister.gov/2016-22379.pdf

the agency to consider working with NLM to add a feature to clinicaltrials.gov such that IND or IDE numbers be provided for trials that are being conducted under an IND or IDE.

In addition to the state laws that govern trade secrets, there are two federal statutes that govern trade secrets that are as follows: the Uniform Trade Secrets Act; and the Defend Trade Secrets Act of 2016. In terms of regulation, 21 CFR § 21.60 governs "[t]rade secrets and commercial or financial information which is privileged or confidential."[9] The regulations codified in the CFR are under FDA's purview to amend, unlike the federal statutes that govern the issue. The drug regulation at 21 CFR § 312.130 states that "(a) The existence of an investigational new drug application will not be disclosed by FDA unless it has previously been publicly disclosed or acknowledged."[10] The parallel regulation for biologics, or 21 CFR § 601.50 (*Confidentiality of data and information in an investigational new drug notice for a biological product*), states the following:[11]

> (a) The existence of an IND notice for a biological product will not be disclosed by the Food and Drug Administration unless it has previously been publicly disclosed or acknowledged.
> (b) The availability for public disclosure of all data and information in an IND file for a biological product shall be handled in accordance with the provisions established in 601.51.
> (c) Notwithstanding the provisions of 601.51, the Food and Drug Administration shall disclose upon request to an individual on whom an investigational biological product has been used a copy of any adverse reaction report relating to such use.

The amending of said regulations may be necessary in order to implement the proposed disclosure requirement of IND/IDE numbers. Many companies already choose to disclose their product candidate names, and it is the opinion of SES that the disclosure of the IND/IDE number is analogous to disclosure of product candidate nomenclature. As long as the INDs and IDEs themselves are still kept confidential by the agency, SES does not anticipate that requiring companies, academic medical centers, *et cetera* to simply disclose IND or IDE numbers as part of their clinicaltrials.gov registration would create major backlash for the agency. [12, 13]

Ensuring that the public be able to access accurate, science-based information regarding HCT/Ps, regardless of whether the products themselves are purview to FDA jurisdiction, should be a high priority of FDA OCTGT, especially where the agency chooses

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[9]Source URL: https://www.gpo.gov/fdsys/pkg/CFR-2012-title21-vol1/xml/CFR-2012-title21-vol1-sec20-61.xml

[10]Source URL: http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm?fr=312.130

[11]Source URL: https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfCFR/CFRSearch.cfm?CFRPart=601&showFR=1&subpartNode=21:7.0.1.1.2.6

[12]See, for example, the following URLs: http://c.ymcdn.com/sites/www.celltherapysociety.org/resource/resmgr/uploads/files/Annual%20Meetings/2011/Presentations/122_1530_Bartido_Ruy_Fri_TAT5.pdf; and http://c.ymcdn.com/sites/www.celltherapysociety.org/resource/resmgr/2013_NARegMeetingPresentations/BARTIDO_Wrkshp2_Edited.pdf.

[13]For market research purposes, SES cautions that the agency should not act to deter foreign companies from registering overseas trials in clinicaltrials.gov as it is the closest site we as industry have to a single data repository for clinical trials.

to exercise regulatory discretion; SES argues that FDA OCTGT should devote less of its resources to the regulation of HCT/Ps that have not been known to cause (serious) adverse events/adverse experiences/unanticipated problems relative to those known to cause serious adverse events, such as chimeric antigen receptor ("CAR") T cell based product candidates regulated by FDA OCTGT under section 351 of the Public Health Service Act ("PHS Act").[14]

B. Homologous Use

As was set forth above, within the framework of the mission statement of FDA, it is the position of SES that FDA OCTGT should devote its limited resources to the regulation of HCT/Ps with known safety issues, such as PHS Act section 351 CAR T cell based product candidates/products. Within this context, SES argues that FDA should exercise regulatory discretion over PHS Act section 361 HCT/Ps based on the risk posed by each product to the public much in the same manner that it differentiates PHS Act section 351 HCT/Ps from PHS Act section 361 HCT/Ps. In other words, a risk based approach, if any, should be considered by FDA if it decides it needs to regulate PHS Act section 361 HCT/Ps in order to protect the public health. As has already been demonstrated, where there is too much regulation, companies may choose to export HCT/P products/product candidates to overseas locations where FDA absolutely lacks jurisdiction. By implementing a common sense, risk based regulatory approach for PHS Act section 361 HCT/Ps, unlike that which would occur if the four draft guidances are implemented as is, FDA could both help to ensure public safety by keeping these HCT/Ps on US soil, where state boards have the authority to revoke medical licensure, while also helping to promote US industry and innovation.

SES also cautions FDA of the unintended consequences of its actions with regard to the four draft guidances. For example, if the draft guidances are implemented as is and fat grafting is no longer an option for breast reconstruction, patients will likely rely further on alternates. In the case of breast reconstruction, implants are an obvious choice in the event that surgeons are no longer allowed to perform fat grafting (without going through the BLA process, which would likely be a huge deterrent for the majority of surgeons), but even the agency itself is not convinced that implants are safe. The FDA's own website includes the following with regard to breast implants:[15]

> In 2011, the FDA identified a possible association between breast implants and the development of anaplastic large cell lymphoma (ALCL), a rare type of non-Hodgkin's lymphoma. At that time, the Agency was aware of approximately 60 cases of ALCL in women with breast implants, out of approximately 5-10 million women who had received breast implants worldwide. This included 34 unique cases that were described in the medical literature from January 1, 1997 through May 21, 2010, and additional cases identified by international regulatory agencies, scientific experts and breast implant manufacturers. Based on this data, the FDA cautioned patients and health care providers that women with breast implants might have a very low but increased risk of developing ALCL.

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[14]SES is the currently the only woman owned small business ("WOSB") operating in the CAR T cell field.

[15]Source URL: http://www.fda.gov/MedicalDevices/ProductsandMedicalProcedures/ImplantsandProsthetics/BreastImplants/ucm239995.htm

Regarding autologous fat grafting for breast reconstruction, why would FDA want to take an option that it is not investigating as a cause of cancer (*i.e.*, fat grafting) off of the table while leaving patients with an option that it is investigating as a possible cause of ALCL (*i.e.,* breast implants)?[16] Regulatory actions (in the form of guidances) that may actually do more harm than good stand in contrast to the mission statement of the agency, and it is within this context that SES cautions FDA of the unintended consequences of the draft guidances if implemented without significant revisions..[17]

C. Additional Recommendations & Clarification

The agency should consider incorporating additional language/amending the language in 21 CFR 1271 such that it gives FDA the option to exercise additional regulatory oversight over certain product categories (*e.g.,* autologous, adipose tissue derived stem cells as well as autologous stem cells obtained from bone marrow) based on the risk/harm profile to patients.[18] For example, the agency may want to consider language which would grant it jurisdictional authority over autologous adipose tissue-derived stem cell injections that have been demonstrated to cause severe vision loss as was reported by Thomas A. Albini, MD at the FDA SEDHC workshop. While the biologics licenses application ("BLA") pathway under PHS Act § 351 may be inappropriate for these types of products, also inappropriate is a complete lack of regulatory oversight where patients are clearly being harmed.

Simply put, regulatory amendments in the CFR should be undertaken such that FDA has the authority to exercise additional regulatory oversight over stem cell based products in cases where harm is clearly being done (*e.g.,* when patients are being blinded) whereas limited federal resources should not be wasted on the oversight of interventions that are not causing (serious) adverse events/adverse experiences/unanticipated problems, whether the treatments are proven to be safe and effective. Based on this premise, FDA should consider incorporating additional language/amending the language in 21 CFR 1271 such that it has the option to exercise additional regulatory oversight over certain product categories that fall within the realm of PHS Act § 361 HCT/Ps in cases of clear patient injury/harm.

III. Conclusion

In summary, for the reasons set forth herein, SES urges the agency to reconsider its proposed regulatory approach to PHS Act section 361 HCT/Ps.

Thank you kindly for the opportunity to submit the comments seen above. As always, please feel free to contact Sara Elizabeth Siegler, CEO/Owner (SES), with any questions and/or concerns regarding said comments.

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[16]See: http://www.fda.gov/MedicalDevices/ProductsandMedicalProcedures/ImplantsandProsthetics/BreastImplants/ucm239995.htm

[17] The CEO/Owner of SES had autologous cartilage grafting done for facial reconstructive purposes.

[18]Whereas FDA lacks the authority to amend the federal laws, such as the PHS Act, the agency does have authority over the federal regulations in the CFR and guidance documents.

APPENDIX

HUMAN CELL AND TISSUE ESTABLISHMENT REGISTRATION (HCTERS) - Public Query Harvard (INACTIVE) 05142016

FDA U.S. Food and Drug Administration Department of Health and Human Services

CENTER FOR BIOLOGICS EVALUATION AND RESEARCH

FDA Home Page | Contact eHCTERS Technical Support

HUMAN CELL AND TISSUE ESTABLISHMENT REGISTRATION - Public Query
Establishment Details

Establishment Name and Location

Current Status:	Inactive
Last Annual Registration Year:	2006
FDA Establishment Identifier (FEI):	3004544669
Establishment Name:	Harvard Gene Therapy Initiative
Address:	4 Blackfan Circle
	Suite 407
City:	Boston
State:	Massachusetts
Zip:	02115
Country:	United States
Phone:	617-432-1465

Establishment Functions

	Types of HCT/P's	Recover	Screen	Test	Package	Process	Store	Label	Distribute
a.	Bone								
b.	Cartilage								
c.	Cornea								
d.	Dura Mater								
e.	Embryo								
f.	Fascia								
g.	Heart Valve								
h.	Ligament								
i.	Oocyte								
j.	Pericardium								
k.	Peripheral Blood Stem Cells								
l.	Sclera								
m.	Semen								
n.	Skin								
o.	Somatic Cell Therapy Products								
p.	Tendon								
q.	Umbilical Cord Blood Stem Cells								
r.	Vascular Graft								

Establishment HCT/P Listing

		HCT/P's	HCT/P's	HCT/P's	

	Types of HCT/P's	Described in 21 CFR 1271.10	Regulated as Medical Devices	Regulated as Drugs or Biological Drugs	Proprietary Names
a.	Bone				
b.	Cartilage				
c.	Cornea				
d.	Dura Mater				
e.	Embryo				
f.	Fascia				
g.	Heart Valve				
h.	Ligament				
i.	Oocyte				
j.	Pericardium				
k.	Peripheral Blood Stem Cells				
l.	Sclera				
m.	Semen				
n.	Skin				
o.	Somatic Cell Therapy Products				
p.	Tendon				
q.	Umbilical Cord Blood Stem Cells				
r.	Vascular Graft				

HCT/P Listing - Donor Information

	Types of HCT/P's	SIP	Directed	Anonymous	Autologous	Family Related	Allogeneic
e.	Embryo						
i.	Oocyte						
k.	Peripheral Blood Stem Cells						
m.	Semen						
o.	Somatic Cell Therapy Products						
q.	Umbilical Cord Blood Stem Cells						

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